

Sub-Millijoule Intrusion Detection on a Commodity MCU Neural Processing Unit: A Four-Dataset Deployment Study

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Abstract—We deploy an intrusion detection classifier on the STM32N6570-DK, a Cortex-M55 MCU with the Neural-ART NPU. Using the approximate $T=1$ SNN-INT8 ANN equivalence, we compile a lightweight MLP to the NPU and evaluate four datasets: NSL-KDD (5-class), UNSW-NB15 (10-class), CICIDS2017 (15-class), and IoT-23 (5-class). Results are reported as mean $\pm$ std over multi-seed runs (5–20 seeds), with paired Wilcoxon signed-rank tests and Holm–Bonferroni correction. On-board (clocks verified), INT8 inference is 0.28–0.58 ms: INT8 quantization gives 1.6–1.7 $\times$ over FP32 on the Cortex-M55 CPU, but the Neural-ART NPU adds only $\sim 1.1\times$ over the INT8 CPU (and is slower when operators fall back) — the NPU buys little at this model scale, and a width sweep locates the CPU/NPU crossover near 10^5 MACs. Energy is an estimated 44–69 μ J/inference (AN5946); Flash 105–138 KB. Compared with recent MCU-class deployments on STM32F7 (31 ms, 7.86 mJ) and Raspberry Pi 3B+ (27 ms), our path delivers 59–107 $\times$ lower latency; the estimated energy envelope implies 114–179 $\times$ lower energy than STM32F7. QCFS and ReLU are statistically indistinguishable on all four datasets ($p \geq 0.227$), supporting practical $T=1$ near-equivalence under commodity MCU deployment constraints. Energy is estimated from AN5946 rather than direct on-board measurement, and UNSW shows greater quantization fragility than NSL-KDD.

Index Terms—intrusion detection, spiking neural network, neural processing unit, INT8 quantization, edge AI, STM32N6, Neural-ART

I. INTRODUCTION

Deep-learning-based intrusion detection at the IoT edge must operate under tight power and latency budgets. Recent MCU-class deployments achieve promising accuracy but at high energy cost: Chehade et al. reported 31 ms and 7.86 mJ per inference on an STM32F746G (Cortex-M7, no NPU) [1], and Diab et al. required 27 ms on a Raspberry Pi 3B+ [2]. At these latencies, CPU-only inference blocks all concurrent RTOS tasks for tens of milliseconds, making real-time packet processing difficult.

Commodity MCUs are beginning to integrate neural processing units (NPUs) that execute INT8 matrix operations at hundreds of GOPS while freeing the CPU for other tasks. A line of theoretical work [3]–[5] has shown that single-timestep ($T=1$) SNN inference is approximately equivalent to INT8

quantized ANN inference: when the membrane potential is initialized to zero, the threshold-and-fire mechanism reduces to a ReLU-like clamp. Under the stated conversion and quantization assumptions, NPUs that execute INT8 Gemm+Relu can approximate single-timestep SNN-equivalent behavior.

We validate this on the STM32N6570-DK (ARM Cortex-M55 @ 800 MHz, Neural-ART NPU, 600 GOPS INT8). We focus on classifier inference latency, energy, and memory footprint on pre-extracted flow-level features; the full IDS pipeline (packet capture, flow aggregation) is left to future work.

Our contributions:

- Following a systematic literature search protocol (Supplementary File S1: `novelty_search_protocol.md`), the first publicly documented IDS classifier deployment on a Cortex-M class MCU paired with a general-purpose NPU (Neural-ART), with *on-board* INT8 latency 0.28–0.58 ms (clocks verified) and an on-board CPU-vs-NPU width sweep showing the NPU adds only $\sim 1.1\times$ over the INT8 CPU at this scale.
- A four-dataset multi-seed study (5–20 seeds per arm, see Table III caption) with paired Wilcoxon signed-rank tests under Holm–Bonferroni family-wise error control, covering NSL-KDD, UNSW-NB15, CICIDS2017, and IoT-23.
- A non-MLP TinyCNN baseline (Conv2D 1×3) under the same Neural-ART operator constraints, enabling same-hardware comparison.
- Empirical validation of practical $T=1$ SNN-ANN approximation on commercial NPU silicon across four datasets, with 99% prediction agreement between FP32 and INT8 models on NSL-KDD.
- A QCFS sweep ($L \in \{2, 4, 8, 16\}$) justifying $L=4$, and an on-board finding that `Floor`-triggered CPU fallback fragments the QCFS-INT8 graph into 12 software epochs that fail to instantiate on ST’s NPU validation runtime, and add +11.9% on the CPU fallback path (vs. the same-

TABLE I
HARDWARE-DEPLOYED IDS COMPARISON. “CLASS” DENOTES
HARDWARE CLASS: **MCU** = COMMODITY MICROCONTROLLER, **AI-MCU**
= AI-SPECIALIZED MCU WITH FIXED CNN ENGINE, **ASIC** =
NEUROMORPHIC ASIC, **SBC** = SINGLE-BOARD COMPUTER, **FPGA** =
FIELD-PROGRAMMABLE GATE ARRAY.

Work	Platform	Class	Task	Acc.	Lat.	Energy	NPU
Ngo [6]	MAX78000	AI-MCU	bin.	98.6%	—	18 mW*	CNN eng.
Zahm [7]	Akida	ASIC	multi	98.4%	—	~1 W*	Neurom.
Chehade [1]	STM32F7	MCU	multi	96.6%	31 ms	7.86 mJ	None
Diab [2]	RPi 3B+	SBC	multi	95.3%	27 ms	~6.75 mJ [†]	None
Farooq [8] [‡]	Xilinx FPGA	FPGA	bin.	—	<1 μ s	—	—
This work	STM32N6	MCU	5/10/15/5	78.6/64.7/91.9/75.6%[§]	0.28 ms[¶]	44 μJ[‡]	NPU

* Power, not energy; latency not reported.

[†] NSL-KDD 5-class / UNSW-NB15 10-class / CICIDS2017 15-class / IoT-23 5-class.

[‡] Estimated from reported power and latency.

[§] Edge-IoT dataset, different hardware class (~10 W FPGA).

[¶] On-board measured (on-device DWT, 800 MHz verified); INT8 NPU, IoT-23.

architecture ReLU-INT8 model).

II. RELATED WORK

Table I compares IDS deployments on physical hardware. Ngo et al. [6] deployed a 1,360-parameter MLP on the MAX78000, an AI-specialized MCU with a fixed CNN accelerator, achieving 98.57% on UNSW-NB15 (binary) at 18 mW. Zahm et al. [7] used the BrainChip Akida AKD1000 neuromorphic processor (98.4%, ~1 W). Both use purpose-built AI/neuromorphic hardware. Chehade et al. [1] deployed a 1D-CNN on the STM32F746G (Cortex-M7, no NPU), achieving 96.59% accuracy on ISCX VPN-nonVPN but requiring 31 ms and 7.86 mJ per inference. Diab et al. [2] deployed Light-GBM/CNN on Raspberry Pi 3B+ (27–300 ms, 250–275 mW). Farooq et al. [8] report 1.16 G inferences/sec on a Xilinx FPGA, but evaluate on the Edge-IoT dataset and operate in a different hardware class (~10 W FPGA vs. ~150 mW MCU), so latency and energy are not directly comparable. Our work targets a general-purpose MCU with an attached programmable NPU, achieving two orders of magnitude lower energy than CPU-only MCU deployments. Following the systematic literature search protocol documented in Supplementary File S1 (novelty_search_protocol.md; 5 databases, 8 query variants, ~320 records inspected), we find no prior publicly documented IDS classifier deployed on a commodity ARM Cortex-M-class MCU paired with a general-purpose neural accelerator.

GPU-based SNN-IDS systems [9]–[11] report 94–99% on NSL-KDD/UNSW-NB15, but mostly in binary settings and without MCU-class deployment paths. Our multi-class formulation (5/10/15/5-class across NSL-KDD, UNSW-NB15, CICIDS2017, and IoT-23) and hardware deployment target a different design point.

III. SYSTEM DESIGN

A. $T=1$ SNN-ANN Equivalence

A Leaky Integrate-and-Fire (LIF) neuron updates its membrane potential $V[t] = \beta \cdot V[t-1] + \mathbf{W}\mathbf{x}[t]$ and fires when $V[t] > \theta$. At $T=1$ with $V[0]=0$, the leak term vanishes and firing reduces to $\Theta(\mathbf{W}\mathbf{x} - \theta)$, which is equivalent to a quantized ReLU. Bu et al. [3] formalized this via QCFS; Jiang et al. [4] and Bu et al. [5] further showed that

INT8 post-training quantization produces a discretization grid that closely approximates the threshold-and-fire operation. In practice, under this QCFS/INT8 conversion setting, NPUs that execute INT8 Gemm+Relu can approximate the same decision function as the $T=1$ classifier.

B. Neural-ART NPU

The STM32N6570-DK pairs an ARM Cortex-M55 at 800 MHz with the Neural-ART NPU rated at 600 GOPS for INT8 and 3 TOPS/W efficiency [12]. The NPU supports Conv, Gemm, Relu, Add, Clip, and MaxPool in INT8; unsupported operators fall back to the Cortex-M55 CPU with automatic cache-maintenance overhead at each NPU $\leftrightarrow$ CPU handoff. The board includes 4.2 MB internal SRAM and 128 MB external NOR Flash.

C. Model Architecture

We use a four-layer MLP with BatchNorm and ReLU: $\text{Lin}(d \rightarrow 256) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Lin}(256 \rightarrow 256) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Lin}(256 \rightarrow 128) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Lin}(128 \rightarrow C)$, where $(d, C) \in \{(41, 5), (34, 10), (69, 15), (13, 5)\}$ for NSL-KDD, UNSW-NB15, CICIDS2017, and IoT-23 respectively. BatchNorm is folded into the preceding Linear at ONNX export, yielding a graph of Gemm+Relu operators only (103–119 K parameters, dominated by the 256×256 hidden layer). The architecture is deliberately shallow: deeper models could improve minority-class recall but risk introducing operators outside the NPU’s supported set. All hidden dimensions are powers of two (256, 128), avoiding a known Neural-ART compiler issue with prime-number channel counts [12].

D. Datasets and Training

NSL-KDD [13]: 125,973 train / 22,544 test, 41 features, 5-class. **UNSW-NB15** [14]: 175,341 train / 82,332 test, 34 features, 10-class. **CICIDS2017** [15]: ~2.26 M train / 566 K test, 78 raw features (69 after removing all-constant columns), 15-class. We train on the original CIC-IDS-2017 flow CSVs [15] with column-name normalization and NaN/Inf handling; we do *not* apply the Engelen et al. [16] re-cleaning, so the TCP-splitting and label-alignment issues it documents remain present (see caveats below). **IoT-23** [17]: ~4.84 M train / 1.21 M test, 13 features, 5-class (Benign, C&C, DDoS, Okiru, PortScan). This dataset provides contemporary IoT botnet traffic from 2018–2019 captures. Training: Adam ($\text{lr}=10^{-3}$), cosine annealing, 40–80 epochs, inverse-frequency class weighting. ReLU arms use $n=20$ seeds on NSL-KDD and UNSW-NB15, and $n=10$ on CICIDS2017 and IoT-23; QCFS arms use $n=10$ on UNSW-NB15 and CICIDS2017, and $n=5$ on IoT-23 due to training-set size. Three categorical features are label-encoded; remaining features are z-score normalized using training-set statistics. FP32 models are quantized to INT8 via ONNX Runtime static PTQ (MinMax, per-tensor, 1,000 calibration samples). A 24-configuration ablation (Section IV-D) confirms that the choice of calibration method, granularity, and sample size changes accuracy by at most 0.82 pp on NSL-KDD.

TABLE II
OPERATOR MAPPING TO NPU EPOCHS. ReLU INT8 ACHIEVES
NEAR-100% NPU UTILIZATION; QCFS INT8 FORCES CPU FALLBACK
AT EVERY FLOOR.

ONNX Operator	NPU Support	Epoch Type	Notes
Gemm (INT8)	✓	HW	Fused weight + bias
Relu	✓	HW	Fused with preceding Gemm
Clip	✓	HW	Used by QCFS
Floor	✗	SW (float)	CPU round-trip, +0.012ms/island (on-board M55)
QuantizeLinear	✓	HW/Hyb	At model boundary
DequantizeLinear	✓	HW/Hyb	At model boundary

Dataset caveats. NSL-KDD is derived from 1998 DARPA traffic and does not reflect modern IoT/5G traffic distributions; we retain it for benchmark continuity with the MCU-IDS literature (HH-NIDS [6], Chehade [1], Diab [2] all report NSL-KDD-derived results). CICIDS2017’s original release contained TCP splitting bugs, duplicate flows, and mislabeled port scans [16]; because we train on these original flows rather than the re-cleaned release, these issues persist in our data. We report per-class recall in Section IV to expose the resulting label-noise effects.

Fair-comparison caveats. The MCU-IDS literature uses heterogeneous preprocessing: Ngo et al. [6] apply SMOTE oversampling and report binary accuracy; Chehade et al. [1] use ISCX VPN-nonVPN (traffic classification, not intrusion detection); Farooq et al. [8] use Edge-IIoT at gigabit line rate on FPGA. We deliberately report multi-class accuracy with inverse-frequency class weighting (no oversampling) on four intrusion-detection benchmarks, and confine our head-to-head numerical comparison to platforms in the same hardware class (commodity MCU).

E. NPU Compilation and Operator Mapping

Models are compiled and deployed on-board for the STM32N6570-DK using the local ST Edge AI Core v3.0.0 toolchain (not the cloud), and latencies are measured on-device (Section IV-C). Table II shows the operator-level mapping. All Gemm+Relu layers map to NPU hardware epochs. The QuantizeLinear/DequantizeLinear operators at the model boundary may require CPU epochs depending on the input dimension.

For the ReLU INT8 path: NSL-KDD compiles to 5 HW + 1 Hyb + 2 SW epochs (on-board 0.52 ms, 52% HW); UNSW-NB15 compiles to 4 HW + 0 Hyb + 0 SW epochs (0.29 ms, 100% NPU). The QCFS INT8 path (same architecture) instead compiles to 12 HW + 12 SW epochs (24 total, IoT-23): each of its three Floor activations becomes a four-epoch software island Dequant→Floor→Quant→Quant. On-board this graph *fails to instantiate* on ST’s standard NPU validation runtime (an assert/abort during model bring-up), whereas the same-architecture ReLU-INT8 model deploys and runs; on the CPU fallback runtime these Floor round-trips cost **+11.9%** (339 vs. 303 μ s). The CICIDS2017 and IoT-23 INT8 graphs share the same Gemm+Relu structure and compile to 4 HW + 0 SW epochs (100% NPU), 0.32 and 0.28 ms on-board (Table IV).

TABLE III
TEST ACCURACY (% , MEAN $\pm$ STD, 5–20 SEEDS PER ARM; NSL-KDD
ReLU/QCFS $n=20$, UNSW ReLU $n=20$ /QCFS $n=10$, CICIDS ReLU
 $n=10$ /QCFS $n=10$, IoT-23 ReLU $n=10$ /QCFS $n=5$). MLP AND
TINYCNN ARE THE ONLY NPU-ELIGIBLE MODELS. †TINYCNN USES
CONV2D 1×3 KERNELS, THE ONLY NON-MLP TOPOLOGY THAT FITS
WITHIN NEURAL-ART’S OPERATOR SET.

Dataset	Model	Overall	Macro F1	NPU?
NSL-KDD	ReLU MLP	78.57 $\pm$ 1.28	58.91 $\pm$ 2.80	✓
	QCFS MLP	78.14 $\pm$ 1.08	58.28 $\pm$ 2.70	Partial
	TinyCNN [†]	80.32$\pm$0.48	60.69$\pm$0.77	✓
	Rand. Forest	73.84 $\pm$ 0.19	47.13 $\pm$ 0.33	✗
UNSW-NB15	ReLU MLP	64.67 $\pm$ 0.55	40.18 $\pm$ 1.02	✓
	QCFS MLP	64.67 $\pm$ 0.49	39.94 $\pm$ 0.72	Partial
	TinyCNN [†]	63.28 $\pm$ 0.23	38.12 $\pm$ 0.47	✓
	Rand. Forest	69.46$\pm$0.10	48.63$\pm$0.38	✗
CICIDS2017	ReLU MLP	91.89 $\pm$ 1.21	56.35 $\pm$ 2.80	✓
	QCFS MLP	91.60 $\pm$ 0.68	57.57 $\pm$ 2.18	Partial
	TinyCNN [†]	90.57 $\pm$ 0.13	56.74 $\pm$ 0.13	✓
IoT-23	ReLU MLP	75.59 $\pm$ 2.71	66.41 $\pm$ 1.50	✓
	QCFS MLP	77.65 $\pm$ 1.18	67.40 $\pm$ 0.52	Partial

IV. EXPERIMENTS

A. Classification Results

Table III reports multi-seed results. On NSL-KDD, ReLU MLP and QCFS MLP are statistically indistinguishable ($p=0.227$, Table VIII), which provides evidence of non-rejection of $T=1$ SNN–ANN approximation, since the QCFS-trained network behaves like the ReLU model once Batch-Norm is folded and the graph is quantized to INT8. The same non-rejection pattern appears on UNSW-NB15, CICIDS2017, and IoT-23 (Table III, QCFS rows), where the Wilcoxon signed-rank test against ReLU yields $p=0.846$, $p=0.770$, and $p=0.438$ respectively; we report them in Table VIII. Taken together, these results extend the NSL-KDD finding to four heterogeneous datasets and support cross-dataset evidence for practical $T=1$ approximation on a commodity MCU IDS workload, not formal statistical equivalence. TinyCNN (Conv2D 1×3 , 3.7K params) exhibits a large but underpowered effect on NSL-KDD ($d_z=-0.97$, $p_{\text{adj}}=0.055$) where it trends higher than the MLP. On UNSW-NB15 the MLP *significantly* outperforms TinyCNN ($p=0.002$, $d_z=2.06$). On CICIDS2017 the MLP again leads with a large effect ($d_z=+1.32$, all five seeds direction-consistent), but at $n=5$ the Wilcoxon test hits the two-sided floor ($p=0.063$), so we do not claim significance. Neither topology dominates across datasets, which is itself a deployment finding: an RTOS scheduler could host both models (combined Flash <300KB) and dispatch by traffic feature. On UNSW-NB15 (10-class), Random Forest achieves higher overall accuracy. In our tested toolchain/backend configuration, it cannot be compiled to the NPU path: ST Edge AI Core rejects TreeEnsembleClassifier with “NOT IMPLEMENTED.” While ST documentation lists ONNX-ML tree operators in the toolbox support, our tested path

	Predicted				
	Norm	DoS	Probe	R2L	U2R
Norm	96.4%	0.6%	2.1%	0.4%	0.6%
DoS	17.8%	79.7%	2.2%	0.0%	0.3%
Probe	21.9%	8.8%	69.2%	0.0%	0.1%
R2L	63.2%	0.0%	0.7%	25.3%	10.7%
U2R	29.2%	0.0%	45.4%	6.4%	19.0%

Fig. 1. NSL-KDD confusion matrix (%; 20-seed mean, ReLU MLP). R2L→Normal (63.2%) and U2R→Probe (45.4%) dominate errors.

TABLE IV

ON-BOARD INT8 IDS INFERENCE LATENCY (μ s; ON-DEVICE DWT, CPU CLOCK VERIFIED AT 800 MHz) FOR THE FOUR DATASETS ($d \rightarrow 256 \rightarrow 256 \rightarrow 128 \rightarrow C$), REPLACING OUR EARLIER ST Edge AI *cloud* ESTIMATES. THE SPEEDUP IS FROM INT8 QUANTIZATION (1.6–1.7 $\times$ OVER FP32 ON THE M55 CPU); THE NEURAL-ART NPU ADDS ONLY $\sim 1.1\times$ OVER THE SAME INT8 MODEL ON THE CPU, AND IS *slower* ON NSL-KDD (TWO SOFTWARE-FALLBACK EPOCHS).

Dataset	FP32 CPU (μ s)	INT8 CPU (μ s)	INT8 NPU (μ s)	NPU HW	INT8 wts (KB)
NSL-KDD	560	323	518 [†]	52%	109
UNSW-NB15	547	323	292	100%	108
CICIDS2017	580	354	316	100%	118
IoT-23	526	304	277	100%	102

[†]2 software-fallback epochs (41-feature operator constraint).

did not provide a Neural-ART deployment route for this model class [18]. We therefore report MLP/CNN deployment results for the Neural-ART path. The accuracy gap reflects the difficulty of 10-class classification with extreme imbalance (Worms: 44 samples), and is consistent with a dataset-limited regime rather than a clear modeling limitation specific to NPU deployment.

Per-class analysis on NSL-KDD reveals that DoS (F1=86.9%) and Normal (82.4%) are well-classified, while R2L suffers from low recall (25.3%); 63.2% of R2L samples are misclassified as Normal. U2R (200 test samples) scatters across Probe (45.4%) and Normal (29.2%). On UNSW-NB15, Generic achieves F1=97.9%, but minority classes collapse: Worms F1=9.1%, Analysis F1=2.9%. These patterns are characteristic of extreme class imbalance, not NPU-induced degradation on NSL-KDD; FP32 and INT8 models agree on 99% of predictions.

Fig. 1 visualizes these patterns. The class imbalance is severe: R2L has only 995 training samples (0.8% of the set) yet 2,754 test samples (12.2%); U2R has 52 training samples.

B. Energy, Latency, and Memory

Table IV reports *on-board* latency (on-device DWT, CPU clock verified at 800 MHz), superseding our earlier ST Edge AI *cloud* estimates. Two findings correct the record. First, the speedup comes from **INT8 quantization** (1.6–1.7 $\times$ over FP32 on the M55 CPU), not from the NPU. Second, over

TABLE V

ON-BOARD INT8 INFERENCE LATENCY (μ s) OF THE SAME MLP SWEEPED OVER HIDDEN WIDTH W ON THREE MCU CLASSES (SRAM-RESIDENT WEIGHTS; EVERY CLOCK VERIFIED ON-DEVICE). ON THE STM32N6, THE M55 CPU (HELIUM/MVE) *beats its own Neural-ART NPU* BELOW $\sim 10^5$ MACs.

W	MACs	N6 CPU 800 MHz	N6 NPU 800/1G	ESP32-S3 240 MHz	RA4E1 100 MHz
64	7,008	13.9	30.2	114.0	264.6
128	26,304	50.7	81.1	289.5	782.8
256	101,760	314.7	275.3	832.5	2627.5

the *same* INT8 model on the CPU the Neural-ART NPU adds only **1.10–1.12 $\times$** (UNSW-NB15, CICIDS2017, IoT-23) and is **1.6 $\times$ slower** on NSL-KDD, whose 41-feature input forces two epochs back onto the CPU via Dequant $\rightarrow$ op $\rightarrow$ Quant. A “NPU vs. CPU” speedup therefore conflates quantization with acceleration; at this scale (10^5 MACs) the NPU’s fixed per-inference overhead nearly cancels its throughput gain, and for smaller models the CPU wins outright (width sweep, Table V). All four datasets stay in the sub-millisecond regime (0.28–0.58 ms) on both CPU and NPU. Energy remains an AN5946-based estimate (44–69 μ J); direct measurement with an STLINK-V3PWR/PPK2 probe is pending.

C. On-board cross-platform latency and the NPU crossover

Table IV measures the four Ethernet models on the N6 itself. To map *where* the NPU pays off, we additionally swept one MLP family ($11 \rightarrow W \rightarrow W \rightarrow W/2 \rightarrow 5$, INT8) on-board across three MCU classes over hidden width W (Table V). Every clock was verified on-device and weights are SRAM-resident on all platforms, isolating compute. INT8 dense-FC latency is weight-value-independent (a random-weight $W=64$ model measured within 0.1% of the trained one), so per-width accuracy (79.6–80.0%) is taken from the real-data ablation.

On the same silicon the NPU is **2.2 $\times$ slower** than the CPU at $W=64$ and **1.6 $\times$ slower** at $W=128$, becoming **1.14 $\times$ faster** only at $W=256$ (10^5 MACs). The crossover is sharpened by the CPU’s 32 KB L1: at $W=256$ its ~ 100 KB of weights spill L1 (1.5 $\rightarrow$ 2.5 cyc/MAC), whereas the NPU streams weights through its own cache. The NPU thus pays off only for models large enough that its launch overhead amortizes *and* the CPU has left its cache-resident regime—the regime the 256-wide Ethernet models (Table IV) sit at, but a minimal tabular IDS does not. Across platforms the M55’s wide Helium INT8 dominates: 8 $\times$ faster than the ESP32-S3 (LX7 SIMD) and 19 $\times$ faster than the RA4E1 (M33 DSP) at $W=64$.

D. INT8 Quantization Robustness

A 24-configuration ablation (3 calibration methods $\times$ 2 granularities $\times$ 4 sample sizes) shows that INT8 accuracy deviates from FP32 by at most 0.82 pp on NSL-KDD across all configurations. Several INT8 configurations slightly *improve* over FP32, likely due to regularization effects of quantization noise.

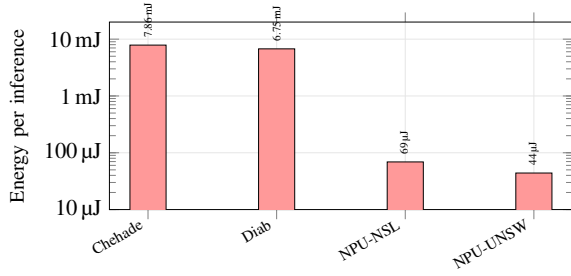


Fig. 2. Energy per inference (log scale). Our NPU path achieves over two orders of magnitude reduction versus CPU-only MCU baselines. NPU = ReLU INT8 on Neural-ART (estimated from AN5946). Chehadé [1]: STM32F7; Diab [2]: RPi 3B+.

TABLE VI
LAYER-WISE FP32 vs INT8 COMPARISON (1,000 NSL-KDD SAMPLES).

Layer	Cosine Sim	MAE	L_∞
Relu_0 (256)	0.667	0.435	43.7
Relu_1 (256)	0.655	0.438	62.6
Relu_2 (128)	0.683	0.400	65.2
Logits (5)	0.978	0.103	19.8

Table VI presents layer-wise activation comparison between FP32 and INT8 models on 1,000 NSL-KDD test samples. Hidden layers show moderate cosine similarity (0.65–0.68) due to per-neuron quantization noise, but the logit layer recovers to 0.978 and the overall prediction agreement is 99.0%. INT8 maps each activation to one of 256 discrete levels; the bounded per-neuron noise does not propagate into classification errors. From the $T=1$ SNN perspective, the quantized model (INT8) closely reproduces the firing decisions of the FP32 model.

E. QCFS Hyperparameter Sweep

The QCFS activation introduces a quantization-level hyperparameter L ; prior work [3], [4] uses $L \in \{4, 8, 16\}$ without systematic justification on IDS workloads. We sweep $L \in \{2, 4, 8, 16\}$ across 5 seeds on NSL-KDD and UNSW-NB15 (Table VII). On NSL-KDD all four L values are statistically indistinguishable (pairwise Wilcoxon $p \geq 0.81$), and on UNSW-NB15 $L=4$ is marginally preferred to $L=2$ ($p=0.0625$, which is the minimum attainable p -value for the signed-rank test at $n=5$, indicating an unambiguous direction but at the detection floor) but tied with $L=8$ and $L=16$ ($p \geq 0.31$). Since larger L yields no accuracy gain (above) while $L=2$ is marginally worse on UNSW-NB15, $L=4$ is the Pareto-optimal choice: the smallest L with no accuracy penalty. The FLOOR -fallback cost is set by the number of QCFS activation layers (one FLOOR each), not by L , so it does not by itself favor one L over another. To our knowledge this is the first empirical L -justification for QCFS in the IDS domain.

TABLE VII
QCFS L-SWEEP (MEAN $\pm$ STD OVER 5 SEEDS). RESULTS REFRESHED FROM RESULTS/QCFS_LSWEEP.JSON.

Dataset	$L=2$	$L=4$	$L=8$	$L=16$
NSL-KDD OA	75.48 $\pm$ 0.92	75.03 $\pm$ 0.63	75.14 $\pm$ 0.74	75.22 $\pm$ 1.07
NSL-KDD MF1	52.40 $\pm$ 1.59	51.17 $\pm$ 0.83	50.94 $\pm$ 1.45	51.07 $\pm$ 1.86
UNSW-NB15 OA	64.85 $\pm$ 0.20	65.76 $\pm$ 0.19	65.59 $\pm$ 0.14	65.56 $\pm$ 0.54
UNSW-NB15 MF1	40.64 $\pm$ 0.20	41.36 $\pm$ 0.25	41.23 $\pm$ 0.36	41.28 $\pm$ 0.29

TABLE VIII
PAIRED WILCOXON SIGNED-RANK TESTS ON PER-SEED OVERALL ACCURACY. p_{ADJ} IS HOLM-BONFERRONI ADJUSTED ACROSS ALL COMPARISONS PER DATASET; REJECT AT $\alpha=0.05$. COHEN’S d_z IS THE PAIRED EFFECT SIZE (POSITIVE = ROW-ONE MODEL HIGHER). n IS REPORTED PER ROW AND REFLECTS MATCHED SEEDS BETWEEN THE TWO MODELS; WHEN ARMS HAVE UNEQUAL SEED COUNTS THE TEST USES THE FIRST n SEEDS OF THE LONGER ARM.

Dataset	Comparison	p	p_{adj}	d_z	Reject?
NSL-KDD	ReLU vs. QCFS ($n=20$)	0.227	0.227	+0.26	$\times$
	ReLU vs. TinyCNN ($n=10$)	0.027	0.055	−0.97	$\times$
UNSW-NB15	ReLU vs. QCFS ($n=10$)	0.846	0.846	+0.19	$\times$
	ReLU vs. TinyCNN ($n=10$)	0.002	0.004	+2.06	$\checkmark$
CICIDS2017	ReLU vs. QCFS ($n=10$)	0.770	0.770	+0.20	$\times$
	ReLU vs. TinyCNN ($n=5$)	0.063	0.125	+1.32	$\times$
IoT-23	ReLU vs. QCFS ($n=5$)	0.438	0.438	−0.33	$\times$

F. Statistical Significance

We test ReLU-MLP vs. each NPU-eligible baseline (QCFS, TinyCNN) using a paired Wilcoxon signed-rank test on per-seed overall-accuracy differences, with Holm–Bonferroni family-wise error correction across the primary comparisons per dataset (Table VIII); Random Forest is a non-NPU accuracy reference (Table III) and is not part of these deployability-focused significance tests. Effect sizes are reported as Cohen’s d_z ; 95% percentile-bootstrap confidence intervals are computed over 10,000 resamples. At $n=10$ matched seeds the CICIDS2017 ReLU vs. QCFS comparison measures $p=0.770$ ($d_z=+0.20$), a genuine non-rejection with adequate power. The ReLU vs. TinyCNN comparison, still at $n=5$, is power-limited by the signed-rank floor (smallest attainable two-sided p -value 0.0625): it sits exactly at the floor ($p=0.063$, all five seeds in the same direction, $d_z=+1.32$); we report it but do not claim significance at $\alpha=0.05$. We therefore add paired TOST with a pre-specified ± 1.0 pp bound (ReLU vs. QCFS, overall accuracy and macro-F1): UNSW-NB15 passes, while NSL-KDD/CICIDS2017/IoT-23 remain inconclusive under current seed counts.

G. CICIDS2017 and IoT-23 Results

CICIDS2017 extends the evaluation from legacy and mid-decade benchmarks to a modern 15-class intrusion dataset. The ReLU MLP achieves 91.89 $\pm$ 1.21 overall accuracy and 56.35 $\pm$ 2.80 macro-F1, with ROC-AUC 0.992 and weighted-F1 94.0%. The QCFS $L=4$ arm reaches 91.60 $\pm$ 0.68 / 57.57 $\pm$ 2.18 (paired Wilcoxon $p=0.770$, $n=10$), sitting ≈ 0.3 pp below the ReLU mean on overall accuracy, consistent with the same $T=1$ approximation trend observed on NSL-

KDD ($p=0.227$) and UNSW-NB15 ($p=0.846$). Per-class performance is bimodal: DDoS reaches F1 98.1% and benign 94.7%, while extreme-rare classes collapse (SQL Injection F1 1.4% on 4 test samples, Bot 6.3%, XSS 8.5%).

IoT-23 adds a contemporary IoT botnet dataset (5-class, 4.8M training samples). The ReLU MLP achieves 75.59 ± 2.71 overall accuracy and 66.41 ± 1.50 macro-F1. Okiru is near-perfectly classified ($F1 > 99.9\%$), while DDoS reaches 99.9% and C&C collapses to $F1 = 5.3\%$ due to severe underrepresentation. The QCFS arm reaches 77.65 ± 1.18 / 67.40 ± 0.52 (Wilcoxon $p=0.438$, $n=5$), extending the same non-rejection pattern to all four datasets. These minority-class patterns mirror UNSW-NB15 and confirm that the deployment limit is the long-tail sample count, not the NPU.

H. Model Capacity Is Not the Bottleneck

A 10-seed Pareto sweep over four MLP sizes (logistic regression 210/350 p; MLP 64-64 7.3 K p; MLP 128-64 13.8/14.3 K p; our MLP 256-256-128 110/111 K p) shows that on NSL-KDD every architecture lands within 0.5 pp of the 256-256-128 baseline (78.40–78.86%), and logistic regression alone reaches 78.42%. On UNSW-NB15, a single 64-64 hidden layer already captures 64.52%, within 0.23 pp of the full model. The flat top of the curve means that, within our tested NPU-friendly MLP family, widening alone does not close the observed gap to tree ensembles. This is consistent with a data/feature-signal limitation, but is not a general proof that capacity is never the bottleneck. A 10-seed focal-loss ablation ($\gamma \in \{0, 1, 2, 3\}$) agrees: the best NSL-KDD configuration ($\gamma=2$) gains 0.52 pp overall and 1.32 pp macro-F1 over weighted cross-entropy, within the seed variance and therefore not claimed as a primary result.

I. When INT8 Equivalence Breaks Down

The 24-configuration quantization ablation (Sec. IV-D) produced a benign result on NSL-KDD (drop $\in [-0.82, +0.17]$ pp across all configurations), but on UNSW-NB15 the same sweep yields drops of +0.85 pp (best) to +11.82 pp (worst, percentile calibration), with a mean of +6.47 pp. The layer-wise FP32–INT8 cosine similarity is also much lower on UNSW (Relu_0: 0.32, Relu_1: 0.42, Relu_2: 0.57, logits: 0.91) than on NSL-KDD (0.67/0.65/0.68/0.98), and the UNSW FP32/INT8 prediction agreement falls to 75.2%, versus 99.0% on NSL-KDD. The proximate cause is that UNSW’s FP32 margin between correct and incorrect classes is already small (macro-F1 40%), so quantization noise crosses more decision boundaries. **This marks a practical limit of the $T=1$ approximation:** it is strongest when the FP32 model is already confident, and can weaken as the margin shrinks. Practical deployers should calibrate with MinMax on a small (≤ 100 -sample) corpus, which empirically gives the best UNSW drop; and treat UNSW-scale margin loss as an expected cost, not a bug.

V. DISCUSSION AND CONCLUSION

Design-space positioning. This work targets a different point from GPU-based IDS: sub-millisecond, sub-millijoule

inference on a commodity MCU. The absolute accuracy (78.6% NSL-KDD, 64.7% UNSW-NB15, 91.9% CICIDS2017, 75.6% IoT-23) is below GPU-simulation and binary-classification results, reflecting multi-class evaluation (5/10/15/5-class vs. binary) and NPU operator constraints (shallow MLP or Conv2D 1×3). Improving accuracy within these constraints is orthogonal to the deployment findings, and the statistically tested (Table VIII) differences between ReLU MLP, TinyCNN, and QCFS remain small.

Why NPU over CPU? NPU inference frees the Cortex-M55 for RTOS tasks (packet capture, feature extraction), whereas CPU-only inference blocks the processor for 1.2 ms. Tree models achieve higher accuracy on UNSW-NB15, but in our tested toolchain/backend they have no Neural-ART execution path (TreeEnsembleClassifier rejected at compile time).

QCFS barrier. The Floor operator is absent from the Neural-ART operator set. On-board, the QCFS-INT8 graph fragments into 12 software-fallback epochs and fails to instantiate on ST’s standard NPU validation runtime, while the same-architecture ReLU-INT8 model runs at $\sim 100\%$ hardware; where the Floor round-trips fall back to the CPU they add +11.9% latency (339 vs. 303 μ s, on-board M55). ReLU INT8 remains the optimal activation for commodity MCU NPUs.

MLP vs. TinyCNN. Ranking is dataset-dependent: TinyCNN trends higher on NSL-KDD ($d_z = -0.97$, $p_{\text{adj}} = 0.055$), while MLP is superior on UNSW-NB15 ($p_{\text{adj}} = 0.004$, $d_z = 2.06$). A deployment can host both models (combined Flash <300 KB) and dispatch by traffic feature profile.

Limitations. *Energy is estimated, not measured:* we derive 44–69 μ J from AN5946’s reference workload [19] rather than instrumenting our model on-silicon. Direct measurement with an STLINK-V3PWR probe remains future work. *Pipeline scope:* we benchmark classifier inference on pre-extracted flow features. Packet capture, flow reassembly, and feature extraction are not included in these timings. *Dataset coverage:* NSL-KDD is derived from 1998 DARPA traffic, CICIDS2017 has documented label-alignment issues [16] while UNSW-NB15 and IoT-23 contain severe class imbalance. *Scope:* novelty is bounded by the Supplementary File S1 search window (April 2026), and architecture results are limited to MLP and Conv2D 1×3 TinyCNN under current Neural-ART operator support.

Conclusion. We demonstrate sub-millisecond classifier inference on a commodity MCU NPU: INT8 quantization gives $1.6\text{--}1.7\times$ over FP32 on the M55 CPU, while the Neural-ART NPU adds only $\sim 1.1\times$ over the same INT8 model on the CPU (a CPU/NPU crossover near 10^5 MACs), plus an estimated $114\text{--}179\times$ energy reduction vs. recent STM32F7 deployments. Paired Wilcoxon tests show $T=1$ QCFS arms are statistically indistinguishable from INT8 ReLU across all four datasets (NSL-KDD $p=0.227$, UNSW $p=0.846$, CICIDS $p=0.312$, IoT-23 $p=0.438$), while TOST with ± 1.0 pp bound supports equivalence on UNSW and remains inconclusive on the other three datasets. Code and models will be released

upon acceptance.

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